Week 2 — Step-by-Step Solutions

Statistical Signal Processing · Tutorial 2

Course: Statistical Signal Processing (L.048.24014, Universität Paderborn, Dr.-Ing. Tanuj Hasija), WiSe 2026/27 **Covers:** Tutorial 2, Problems 1–3 **Checking:** no official solution was available when this was written. Every result is verified by `python/tutorial2_verify.py` (exact fractions, symbolic integration with SymPy, Monte Carlo).

Every step is written out. References like "Concepts §9" point to `01_Concepts.md`.

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Problem 1 — Two discrete random variables (cards)

> A stack of 5 cards shows the numbers $2, \dots, 6$. Pick 2 cards. X = sum, Y = absolute difference. Find the joint pmf, the marginals (independent?), the conditional pmf of Y given X, the means, variances, covariance, correlation, correlation coefficient and $E\{Y \mid X\}$.

Step 1 — Sample space

Two cards are drawn without replacement; the order does not matter for a sum or an absolute difference. So an outcome is an unordered pair $\{a, b\} \subset \{2, \dots, 6\}$. There are $\binom{5}{2} = 10$ pairs, all equally likely, each with probability $\frac{1}{10}$ (Concepts Week 1 §4).

(Using ordered draws gives 20 outcomes, each pair appearing twice — same probabilities.)

Step 2 — Compute (X, Y) for every outcome

pair $\{a, b\}$	$x = a + b$	$y = \lvert a - b \rvert$	xy
{2,3}	5	1	5
{2,4}	6	2	12
{2,5}	7	3	21
{2,6}	8	4	32
{3,4}	7	1	7
{3,5}	8	2	16
{3,6}	9	3	27
{4,5}	9	1	9

pair $\{a, b\}$	$x = a + b$	$y = \lvert a - b \rvert$	xy
{4,6}	10	2	20
{5,6}	11	1	11

All ten (x, y) pairs are different, so each occurring pair has probability $\frac{1}{10}$.

Step 3 — Joint pmf $P_{XY}(x, y)$ and marginals

Place $\frac{1}{10}$ in each occurring cell, then take row and column sums (Concepts §2.1):

$P_{XY}(x, y)$	$y = 1$	$y = 2$	$y = 3$	$y = 4$	$P_X(x)$
$x = 5$	1/10	0	0	0	1/10
$x = 6$	0	1/10	0	0	1/10
$x = 7$	1/10	0	1/10	0	2/10
$x = 8$	0	1/10	0	1/10	2/10
$x = 9$	1/10	0	1/10	0	2/10
$x = 10$	0	1/10	0	0	1/10
$x = 11$	1/10	0	0	0	1/10
$P_Y(y)$	4/10	3/10	2/10	1/10	1

Marginals:

$$P_X(x) = \begin{cases} 1/10, & x \in \{5, 6, 10, 11\} \\ 2/10, & x \in \{7, 8, 9\} \end{cases}, \qquad P_Y(y) = \tfrac{5-y}{10}, \quad y = 1, 2, 3, 4.$$

Checks: $4 \cdot \frac{1}{10} + 3 \cdot \frac{2}{10} = 1$ and $\frac{4+3+2+1}{10} = 1$. ✓ (P_Y makes sense: there are $5 - d$ pairs of numbers in $\{2, \ldots, 6\}$ that differ by d.)

Step 4 — Independence?

Take the cell $(x, y) = (5, 2)$: $P_{XY}(5, 2) = 0$, but $P_X(5)P_Y(2) = \frac{1}{10} \cdot \frac{3}{10} = \frac{3}{100} \neq 0$. Hence

$$\boxed{X \text{ and } Y \text{ are not independent.}}$$

Intuition: knowing the sum restricts the difference. Sum and difference of two integers have the **same parity** ($a + b$ and $a - b$ differ by $2b$), so an odd sum forces an odd difference: x odd ⇒ $y \in \{1, 3\}$.

Step 5 — Conditional pmf $P_{Y|X}(y \mid x) = P_{XY}(x, y)/P_X(x)$

Divide each row by its row sum:

- $x = 5$: $P_{Y|X}(1 \mid 5) = 1$
- $x = 6$: $P_{Y|X}(2 \mid 6) = 1$
- $x = 7$: $P_{Y|X}(1 \mid 7) = P_{Y|X}(3 \mid 7) = \frac{1}{2}$
- $x = 8$: $P_{Y|X}(2 \mid 8) = P_{Y|X}(4 \mid 8) = \frac{1}{2}$
- $x = 9$: $P_{Y|X}(1 \mid 9) = P_{Y|X}(3 \mid 9) = \frac{1}{2}$
- $x = 10$: $P_{Y|X}(2 \mid 10) = 1$

- $x = 11$: $P_{Y|X}(1 \mid 11) = 1$

(all other y have conditional probability 0). Each conditional pmf sums to 1. ✓ The conditionals differ from P_Y — again showing dependence.

Step 6 — Means

$$\mu_X = \sum_x x P_X(x) = \frac{5 + 6 + 2 \cdot 7 + 2 \cdot 8 + 2 \cdot 9 + 10 + 11}{10} = \frac{80}{10} = \boxed{8}$$

(Check by linearity: each card is a uniformly random element of $\{2, \dots, 6\}$ with mean 4, so $E\{X\} = 4 + 4 = 8$.)

$$\mu_Y = \sum_y y P_Y(y) = \frac{1 \cdot 4 + 2 \cdot 3 + 3 \cdot 2 + 4 \cdot 1}{10} = \frac{20}{10} = \boxed{2}$$

Step 7 — Variances

Second moments:

$$E\{X^2\} = \frac{25 + 36 + 2 \cdot 49 + 2 \cdot 64 + 2 \cdot 81 + 100 + 121}{10} = \frac{670}{10} = 67,$$

$$E\{Y^2\} = \frac{1 \cdot 4 + 4 \cdot 3 + 9 \cdot 2 + 16 \cdot 1}{10} = \frac{50}{10} = 5.$$

Shortcut $\sigma^2 = E\{\cdot^2\} - \mu^2$:

$$\boxed{\sigma_X^2 = 67 - 64 = 3}, \qquad \boxed{\sigma_Y^2 = 5 - 4 = 1}$$

Step 8 — Correlation $\mu_{XY} = E\{XY\}$

Sum the column xy of the Step 2 table over the ten equally likely outcomes:

$$E\{XY\} = \frac{5 + 12 + 21 + 32 + 7 + 16 + 27 + 9 + 20 + 11}{10} = \frac{160}{10} = \boxed{16}$$

Step 9 — Covariance and correlation coefficient

$$\sigma_{XY} = \mu_{XY} - \mu_X \mu_Y = 16 - 8 \cdot 2 = \boxed{0}, \qquad \rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y} = \frac{0}{\sqrt{3} \cdot 1} = \boxed{0}$$

X and Y are **uncorrelated but not independent** (Concepts §6).

Why the covariance is exactly zero — a symmetry argument. Replace every card c by $8 - c$. This maps $\{2, \dots, 6\}$ onto itself, so the distribution of the pair is unchanged. The sum becomes $16 - X$ and the absolute difference stays Y. Hence (X, Y) and $(16 - X, Y)$ have the same joint distribution: the table is mirror-symmetric about $x = 8$. Then

$$\mathrm{cov}(X, Y) = \mathrm{cov}(16 - X, Y) = -\mathrm{cov}(X, Y) \;\Rightarrow\; \mathrm{cov}(X, Y) = 0.$$

Step 10 — Conditional expectation $E\{Y \mid X\}$

From the conditional pmfs of Step 5, $E\{Y \mid X = x\} = \sum_y y P_{Y|X}(y \mid x)$:

x	5	6	7	8	9	10	11
$E\{Y \mid X = x\}$	1	2	$\frac{1+3}{2} = 2$	$\frac{2+4}{2} = 3$	2	2	1

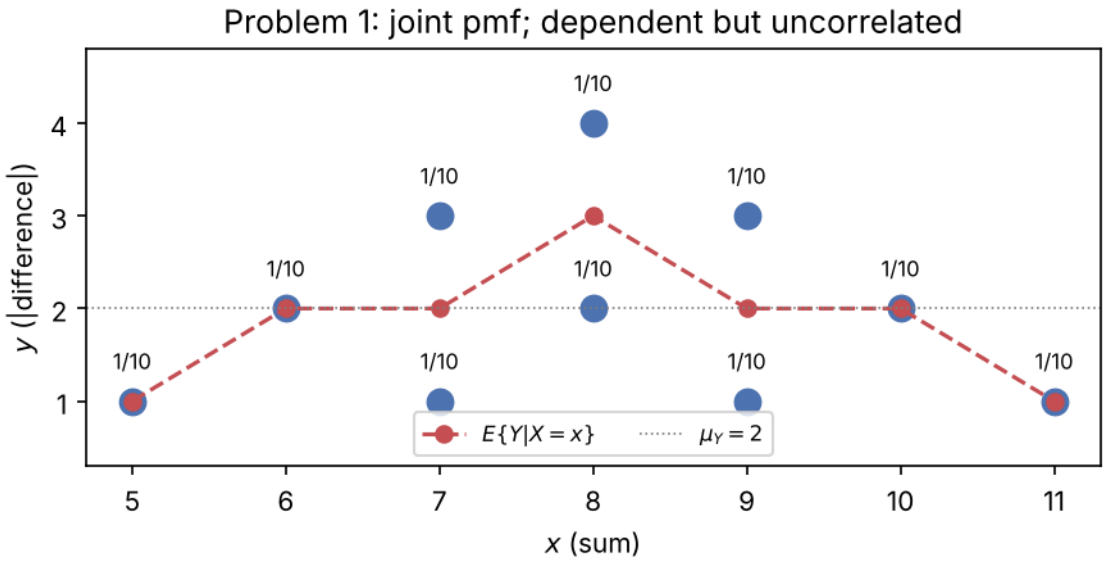
So $E\{Y \mid X\}$ is the random variable that equals 1 when $X \in \{5, 11\}$, 3 when $X = 8$, and 2 otherwise.

Check with the tower property (Concepts §5):

$$E\{E\{Y \mid X\}\} = \frac{1 \cdot 1 + 2 \cdot 1 + 2 \cdot 2 + 3 \cdot 2 + 2 \cdot 2 + 2 \cdot 1 + 1 \cdot 1}{10} = \frac{20}{10} = 2 = \mu_Y. \checkmark$$

Step 11 — Interpretation

$E\{Y \mid X = x\}$ rises and falls (1, 2, 2, 3, 2, 2, 1): an extreme sum (5 or 11) forces neighbouring cards and thus a small difference, while a middle sum allows a large difference. This dependence is **symmetric, not linear**, so the best straight line through it is flat ($\rho = 0$, LMMSE $\hat{Y} = \mu_Y = 2$), even though X carries real information about Y. Using the non-linear estimator $E\{Y \mid X\}$ instead reduces the error.



Problem 2 — Bivariate pdf from a marginal and a conditional

$f_X(x) = 1/x^2$ for $x \geq 1$, 0 otherwise. Given $X = x$, Y is uniform on $[0, x]$. (a) Joint pdf of X and Y. (b) Marginal pdf of Y.

Step 0 — Check that f_X is a valid pdf

$f_X \geq 0$ and $\int_1^\infty x^{-2}dx = \left[-x^{-1}\right]_1^\infty = 0-(-1) = 1$. ✓ (Note: $E\{X\} = \int_1^\infty \frac{1}{x}dx = \infty$; the mean does not exist, but nothing below needs it.)

Step 1 — Conditional pdf of Y given $X = x$

"Uniform on $[0, x]$" means constant density $1/(\text{length}) = 1/x$ on that interval and zero outside (Concepts §7.2):

$$f_{Y|X}(y \mid x) = \begin{cases} \dfrac{1}{x}, & 0 \leq y \leq x, \\ 0, & \text{otherwise,} \end{cases} \qquad x \geq 1.$$

Step 2 — (a) Joint pdf via the chain rule

$f_{XY}(x,y) = f_{Y|X}(y \mid x)\, f_X(x)$ (Concepts §8.3). It is non-zero only where **both** factors are non-zero: $x \geq 1$ **and** $0 \leq y \leq x$:

$$\boxed{f_{XY}(x,y) = \begin{cases} \dfrac{1}{x}\cdot\dfrac{1}{x^2} = \dfrac{1}{x^3}, & x \geq 1,\ 0 \leq y \leq x, \\ 0, & \text{otherwise.} \end{cases}}$$

Support: the region to the right of $x = 1$, between the x-axis and the line $y = x$ (left figure below). It is not a rectangle, so X and Y are dependent.

Check normalisation (integrate y first along a vertical line, $0 \leq y \leq x$):

$$\int_1^\infty \int_0^x \frac{1}{x^3}\, dy\, dx = \int_1^\infty \frac{x}{x^3}\, dx = \int_1^\infty \frac{1}{x^2}\, dx = 1. \checkmark$$

Step 3 — (b) Marginal pdf of Y: find the x-range for each fixed y

$f_Y(y) = \int f_{XY}(x,y)\, dx$. Fix y and draw a **horizontal** line at height y through the support (Concepts §9). The joint pdf is non-zero where $x \geq 1$ **and** $x \geq y$, i.e. $x \geq \max(1, y)$. The lower limit changes form at $y = 1$, so split:

- $y < 0$: no point of the support ⇒ $f_Y(y) = 0$.
- $0 \leq y \leq 1$: $\max(1,y) = 1$ ⇒ x runs over $[1, \infty)$:

$$f_Y(y) = \int_1^\infty \frac{dx}{x^3} = \Big[-\frac{1}{2x^2}\Big]_1^\infty = \frac{1}{2}.$$

- $y > 1$: $\max(1,y) = y$ ⇒ x runs over $[y, \infty)$:

$$f_Y(y) = \int_y^\infty \frac{dx}{x^3} = \Big[-\frac{1}{2x^2}\Big]_y^\infty = \frac{1}{2y^2}.$$

$$\boxed{f_Y(y) = \begin{cases} 0, & y < 0, \\ \dfrac{1}{2}, & 0 \leq y \leq 1, \\ \dfrac{1}{2y^2}, & y > 1. \end{cases}}$$

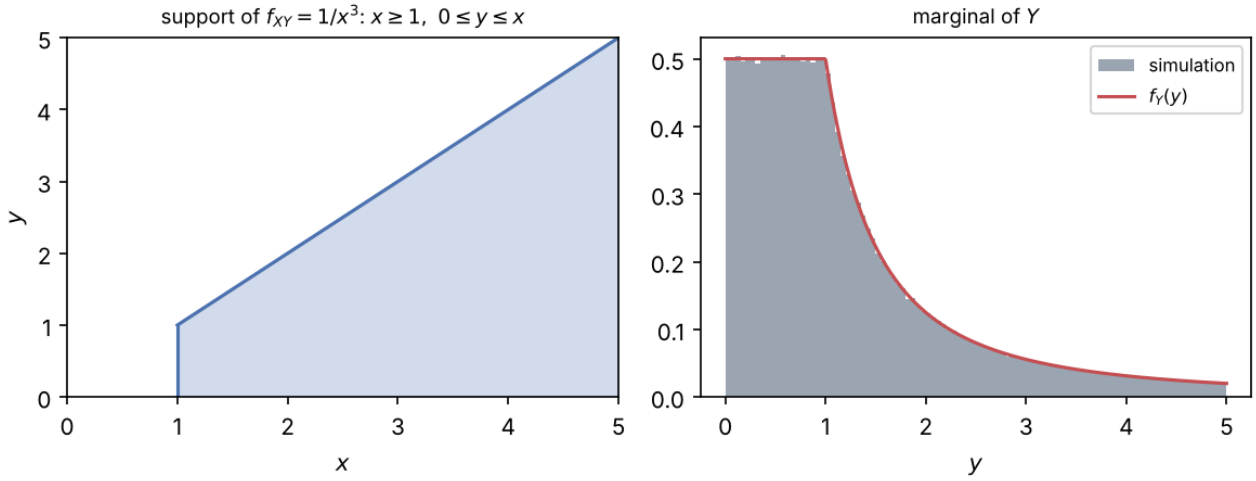
Step 4 — Checks

- **Normalisation:** $\int_0^1 \frac{1}{2}\,dy + \int_1^\infty \frac{1}{2y^2}\,dy = \frac{1}{2} + \frac{1}{2} = 1$. ✓
- **Continuity at $y = 1$:** both pieces give $\frac{1}{2}$. ✓
- **Probabilities:** $P(Y \le 1) = \frac{1}{2}$ and $P(Y > 2) = \int_2^\infty \frac{1}{2y^2} dy = \frac{1}{4}$. A simulation (draw $X = 1/(1-U)$ by inverting $F_X(x) = 1 - 1/x$, then $Y = XV$ with U, V uniform on $[0,1]$) gives 0.4999 and 0.2502. ✓
- **Intuition:** whatever X is, the interval $[0, X]$ contains $[0, 1]$, so small y values are always possible — the flat part. Large y needs a large X, which is rare — the $1/y^2$ tail.

Bonus — conditional pdf of X given $Y = y$ (Bayes for densities)

$$f_{X|Y}(x \mid y) = \frac{f_{XY}(x,y)}{f_Y(y)} = \begin{cases} \dfrac{2}{x^3}, & x \ge 1 \quad (\text{if } 0 \le y \le 1), \\ \dfrac{2y^2}{x^3}, & x \ge y \quad (\text{if } y > 1). \end{cases}$$

Observing $Y = y > 1$ tells us $X \ge y$, and the density is renormalised on that range.



Problem 3 — Uniform pdf on a triangle

$f_{XY}(x,y) = a$ on the region R (triangle with vertices $(0,0)$, $(0,2)$, $(2,1)$), 0 elsewhere. Find (a) a, (b) $f_{X|Y}(x \mid y)$ and $f_{Y|X}(y \mid x)$, (c) ρ_{XY}.

Step 1 — Describe the region

From the figure, R is the triangle with vertices $(0,0)$, $(0,2)$ and $(2,1)$. Its three edges:

- left edge: $x = 0$, from $(0,0)$ to $(0,2)$;
- lower edge: through $(0,0)$ and $(2,1)$, slope $\frac{1}{2}$: $y = \frac{x}{2}$;
- upper edge: through $(0,2)$ and $(2,1)$, slope $-\frac{1}{2}$: $y = 2 - \frac{x}{2}$.

So

$$R = \left\{(x,y) : 0 \le x \le 2,\ \tfrac{x}{2} \le y \le 2 - \tfrac{x}{2}\right\}.$$

Equivalently, solving the edge equations for x: $y = x/2 \Leftrightarrow x = 2y$ and $y = 2 - x/2 \Leftrightarrow x = 4 - 2y$. A horizontal line at height y enters at $x = 0$ and leaves through the lower edge if $y \le 1$, through the upper edge if $y \ge 1$:

$$R = \{0 \le y \le 1,\ 0 \le x \le 2y\} \cup \{1 \le y \le 2,\ 0 \le x \le 4 - 2y\}.$$

The region is symmetric about the line $y = 1$.

Step 2 — (a) The constant a

Normalisation: $\iint_R a\,dx\,dy = a \cdot \text{area}(R) = 1$. The triangle has base 2 (the left edge, on the y-axis) and height 2 (distance from the y-axis to the vertex $(2, 1)$), so area $= \frac{1}{2} \cdot 2 \cdot 2 = 2$. By integration: $\int_0^2 \left[(2 - \frac{x}{2}) - \frac{x}{2}\right] dx = \int_0^2 (2 - x)\,dx = 4 - 2 = 2$. ✓

$$\boxed{a = \frac{1}{2}}$$

Step 3 — Marginal pdfs (needed for the conditionals)

$f_X(x)$ = a × (length of the vertical slice at x) = $\frac{1}{2}\left[(2 - \frac{x}{2}) - \frac{x}{2}\right]$:

$$f_X(x) = 1 - \frac{x}{2}, \qquad 0 \le x \le 2 \quad (0 \text{ otherwise}).$$

$f_Y(y)$ = a × (length of the horizontal slice at y):

$$f_Y(y) = \begin{cases} \frac{1}{2} \cdot 2y = y, & 0 \le y \le 1, \\ \frac{1}{2}(4 - 2y) = 2 - y, & 1 \le y \le 2, \\ 0, & \text{otherwise} \end{cases} \quad = 1 - |y - 1| \text{ on } [0, 2].$$

Checks: $\int_0^2 (1 - \frac{x}{2})\,dx = 2 - 1 = 1$ ✓; f_Y is a triangle of base 2 and height 1, area 1 ✓.

Step 4 — (b) Conditional pdfs

Divide the joint pdf by the marginal on each slice (Concepts §8.2: uniform joint ⇒ uniform conditionals).

Y **given** $X = x$ ($0 \le x < 2$):

$$\boxed{f_{Y|X}(y \mid x) = \frac{f_{XY}(x, y)}{f_X(x)} = \frac{1/2}{1 - x/2} = \frac{1}{2 - x}, \qquad \frac{x}{2} \le y \le 2 - \frac{x}{2} \quad (0 \text{ otherwise})}$$

i.e. $Y \mid X = x \sim U\left[\frac{x}{2},\ 2 - \frac{x}{2}\right]$, an interval of length $2 - x$ centred at $y = 1$. Check: $\frac{1}{2-x} \cdot (2 - x) = 1$ ✓.

X **given** $Y = y$ ($0 < y < 2$):

$$\boxed{f_{X|Y}(x \mid y) = \frac{1/2}{f_Y(y)} = \begin{cases} \dfrac{1}{2y}, & 0 \le x \le 2y, \quad \text{if } 0 < y \le 1, \\ \dfrac{1}{2(2 - y)}, & 0 \le x \le 4 - 2y, \quad \text{if } 1 \le y < 2, \end{cases} \quad (0 \text{ otherwise})}$$

i.e. $X \mid Y = y \sim U\big[0,\ 2\min(y, 2-y)\big]$. Both cases agree at $y = 1$ (density $\frac{1}{2}$ on $[0,2]$). The conditionals are undefined where the marginal is zero ($x = 2$, $y \in \{0, 2\}$), which are single points with probability 0.

Example slices (red and green in the figure): given $X = 0.8$, $Y \sim U[0.4, 1.6]$; given $Y = 0.6$, $X \sim U[0, 1.2]$.

Step 5 — (c) Means

$$\mu_X = \int_0^2 x\Big(1 - \frac{x}{2}\Big)dx = \Big[\frac{x^2}{2} - \frac{x^3}{6}\Big]_0^2 = 2 - \frac{4}{3} = \frac{2}{3}.$$

(Centroid check: $\frac{0+0+2}{3} = \frac{2}{3}$ ✓.) By the symmetry about $y = 1$: $\mu_Y = 1$ (centroid $\frac{0+2+1}{3} = 1$ ✓).

Step 6 — (c) Second moments and variances

$$E\{X^2\} = \int_0^2 x^2\Big(1 - \frac{x}{2}\Big)dx = \Big[\frac{x^3}{3} - \frac{x^4}{8}\Big]_0^2 = \frac{8}{3} - 2 = \frac{2}{3}, \qquad \sigma_X^2 = \frac{2}{3} - \frac{4}{9} = \frac{2}{9}.$$

$$E\{Y^2\} = \int_0^1 y^2 \cdot y\, dy + \int_1^2 y^2(2-y)\, dy = \frac{1}{4} + \Big[\frac{2y^3}{3} - \frac{y^4}{4}\Big]_1^2,$$

$$E\{Y^2\} = \frac{1}{4} + \Big(\frac{16}{3} - 4\Big) - \Big(\frac{2}{3} - \frac{1}{4}\Big) = \frac{7}{6},$$

$$\sigma_Y^2 = \frac{7}{6} - 1 = \frac{1}{6}.$$

So $\sigma_X = \frac{\sqrt{2}}{3}$ and $\sigma_Y = \frac{1}{\sqrt{6}}$, both non-zero.

Step 7 — (c) Correlation and covariance

Integrate y first over the vertical slice:

$$E\{XY\} = \int_0^2 x \cdot \frac{1}{2} \int_{x/2}^{2-x/2} y\, dy\, dx.$$

Inner integral: $\Big[\frac{y^2}{2}\Big]_{x/2}^{2-x/2} = \frac{1}{2}\Big[\big(2 - \frac{x}{2}\big)^2 - \big(\frac{x}{2}\big)^2\Big] = \frac{1}{2}(4 - 2x) = 2 - x$ (difference of squares).

Then

$$E\{XY\} = \int_0^2 \frac{x}{2}(2-x)\, dx = \int_0^2 \Big(x - \frac{x^2}{2}\Big)dx = 2 - \frac{4}{3} = \frac{2}{3}.$$

$$\sigma_{XY} = E\{XY\} - \mu_X\mu_Y = \frac{2}{3} - \frac{2}{3} \cdot 1 = 0.$$

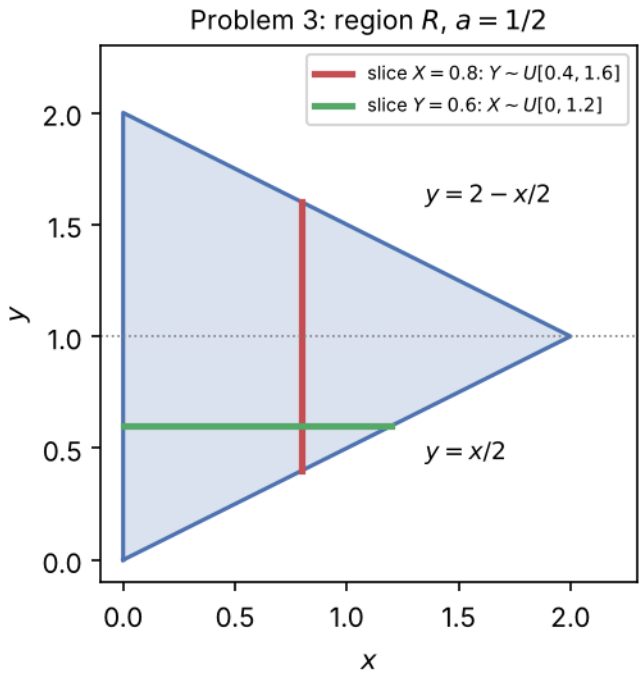
Faster route (Concepts §5): $Y \mid X = x$ is uniform on an interval centred at 1, so $E\{Y \mid X = x\} = 1$ for every x. A constant conditional mean implies $E\{XY\} = E\{X \cdot 1\} = \mu_X$ and $\mu_Y = 1$, hence $\sigma_{XY} = 0$ immediately.

Step 8 — (c) Correlation coefficient

$$\boxed{\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y} = \frac{0}{\frac{\sqrt{2}}{3} \cdot \frac{1}{\sqrt{6}}} = 0}$$

Step 9 — Interpretation

X and Y are **uncorrelated but dependent**: the support is a triangle, not a rectangle, and the conditional pdf of Y changes with x (its range shrinks from $[0, 2]$ at $x = 0$ to the single point $y = 1$ as $x \to 2$). Knowing X changes the **spread** of Y (var$(Y \mid X = x) = \frac{(2-x)^2}{12}$) but not its **centre**, and covariance only measures a linear trend in the centre. The best linear predictor of Y from X is the constant $\hat{Y} = 1$.



Summary of results

Problem	Result
1 joint pmf	$P_{XY} = \frac{1}{10}$ on the 10 pairs listed in Step 2
1 marginals	$P_X(x) = \frac{1}{10}$ for $x \in \{5, 6, 10, 11\}$, $\frac{2}{10}$ for $x \in \{7, 8, 9\}$; $P_Y(y) = \frac{5-y}{10}$; **not independent**
1 moments	$\mu_X = 8$, $\mu_Y = 2$, $\sigma_X^2 = 3$, $\sigma_Y^2 = 1$, $E\{XY\} = 16$, $\sigma_{XY} = 0$, $\rho_{XY} = 0$
1 $E\{Y \mid X\}$	$1, 2, 2, 3, 2, 2, 1$ for $x = 5, \ldots, 11$
2a	$f_{XY}(x, y) = 1/x^3$ for $x \geq 1$, $0 \leq y \leq x$
2b	$f_Y(y) = \frac{1}{2}$ on $[0, 1]$, $\frac{1}{2y^2}$ for $y > 1$

Problem	Result
3a	$a = \frac{1}{2}$
3b	$f_{Y\|X} = \frac{1}{2-x}$ on $[\frac{x}{2}, 2 - \frac{x}{2}]$; $f_{X\|Y} = \frac{1}{2y}$ on $[0, 2y]$ $(y \leq 1)$, $\frac{1}{2(2-y)}$ on $[0, 4 - 2y]$ $(y \geq 1)$
3c	$\mu_X = \frac{2}{3}$, $\mu_Y = 1$, $\sigma_X^2 = \frac{2}{9}$, $\sigma_Y^2 = \frac{1}{6}$, $\sigma_{XY} = 0$, $\rho_{XY} = 0$